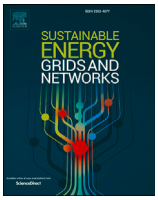
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Fast security-constrained optimal power flow under topology variations via heterogeneous graph neural network learning

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ABSTRACT

The increasing penetration of renewable generation and the rising variability of dynamic loads in modern power systems pose significant challenges for real-time optimization. To address these challenges, we propose a data-driven, physics-informed heterogeneous graph neural network (PIHGNN) framework. The framework learns offline mappings between power system grid topologies and heterogeneous graph neural networks (HGNN), accounting for the distinct physical roles of generator and load nodes, enabling efficient real-time optimal power flow (OPF) calculations under varying grid topologies. Case studies on the IEEE 39-bus and 118-bus systems demonstrate significant computational speedups of ×168.21 and ×218.85, respectively, compared with traditional optimization solvers under ±10% load deviations. The method achieves high-quality solutions with average cost deviations of 0.1276% and 0.0282% for the respective systems relative to MATPOWER benchmarks, while maintaining a perfect feasibility rate of 100% for both systems. Robustness analyses under N-1 topology changes further confirm practical applicability, with feasibility reaching 100% across diverse topology change scenarios for the larger system. Results also demonstrate strong scalability, as the OPF computation performance remains stable with increasing power system size, consistently achieving sub-millisecond solution times. Convergence rates of 100%, combined with perfect feasibility, underscore the method's reliability for real-time dispatch. Overall, PIHGNN offers an effective topology-aware feature extraction mechanism for fast and scalable power system optimization, with future work focused on incorporating stricter physical constraint enforcement and online learning to further enhance robustness and adaptability.

1. Introduction

Optimal Power Flow (OPF) is a cornerstone optimization task in power system operation. It determines operating setpoints—e.g., generator dispatch and bus voltage levels (and, for AC-OPF, voltage angles and reactive power)—to minimize an objective such as generation cost or network losses, subject to AC network physics and operational constraints [1]. However, power systems are rapidly evolving from predominantly centralized control toward more distributed, renewable-rich architectures [2,3]. Although renewable energy sources (RESs) deliver clear environmental benefits, their intermittency and forecast uncertainty introduce pronounced variability into system operation [4]. To explicitly account for such uncertainty, operators often adopt stochastic OPF (S-OPF) formulations, commonly implemented via scenario-based approaches that require solving a large number (often thousands) of OPF instances [5,6]. This creates a key computational bottleneck: while conventional numerical solvers are accurate, their per-scenario runtime accumulates into substantial latency. For example, solving thousands of AC-OPF instances for scenario analysis can take on the order of tens of minutes, which is incompatible with sub-second decision cycles required for real-time dispatch and fast N-1 contingency assessment [7].

Conventional numerical optimization approaches for AC-OPF—such as interior-point methods (IPMs) and sequential quadratic programming (SQP)—are highly effective on many small- and medium-scale instances, but can encounter scalability and robustness challenges as network size grows and the problem becomes strongly non-convex [8–10]. In renewable-rich settings, where operating points change rapidly and many scenarios or contingencies must be evaluated, these solvers may become computationally prohibitive, and their convergence behavior can degrade in high-dimensional or ill-conditioned cases [11,12]. This motivates data-driven surrogates that shift the computational burden

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to an offline training stage and enable millisecond-level inference for online decision support.

Machine learning (ML) has attracted significant interest as a potential remedy to OPF scalability limitations [13]. A range of surrogate architectures has been explored, including reservoir computing for stability-basin inference [14], convolutional neural networks (CNNs) such as ConvOPF that learn mappings from loading conditions to dispatch-related variables [15], and multilayer perceptrons (MLPs) for end-to-end prediction of OPF solutions [16–18]. To improve physical interpretability and constraint awareness, physics-guided variants—e.g., stacked extreme learning machines (SELM) [19] and physics-informed neural networks (PINNs) [20]—have also been investigated. However, a recurring limitation of many conventional ML surrogates is that they represent the grid by flattening system information into fixed-length feature vectors, which largely obscures the underlying network connectivity and adjacency structure that are fundamental to graph-structured power systems [21].

Given the intrinsically graph-structured nature of power grids, graph neural networks (GNNs) provide a natural modeling paradigm by representing buses as nodes and transmission lines as edges [22]. Their neighborhood-aggregation mechanism is well suited to a range of grid-learning tasks, including power-flow approximation [23–25], state estimation [26,27], and contingency screening. However, two limitations hinder the direct use of standard GNNs for OPF surrogates. First, as message-passing depth increases, *over-smoothing* can cause node embeddings to become increasingly similar, weakening the model's ability to preserve heterogeneous electrical roles (e.g., generator versus load buses) [28]. Second, conventional GNNs do not explicitly enforce system-level physical constraints—such as Kirchhoff-consistent nodal power balance and operational limits—which may produce predictions that are computationally efficient yet physically infeasible [29].

However, many existing data-driven OPF surrogates either (i) ignore the grid's graph structure by flattening topology into fixed-length feature vectors, or (ii) rely on homogeneous message passing that can blur fundamentally different electrical roles, and they often lack an explicit mechanism to incorporate system-wide physical constraints. To address these gaps, we propose a Physics-Informed Heterogeneous Graph Neural Network (PIHGNN). PIHGNN represents the grid as a heterogeneous graph by separating generator nodes (PV buses and the slack bus) from load nodes (PQ buses), and by using relation-specific message passing over two edge types: Transmits edges connecting generators to load buses and Connects edges linking load buses. In addition, a physics-informed training objective penalizes violations of AC power-balance relations and operational limits, which empirically yields highly feasible solutions in our experiments. Extensive evaluations on the IEEE 39-bus and IEEE 118-bus systems show that PIHGNN delivers speedups of ×168.21 and ×218.85 over MATPOWER, respectively, while maintaining near-optimal economic performance with mean cost deviations of $\delta_C = 0.1276\%$ (39-bus) and $\delta_C = 0.0282\%$ (118-bus), i.e., well below 0.15%. These results position PIHGNN as a practical, low-latency surrogate for time-critical AC-OPF decision support.

The remainder of this paper is organized as follows. Section 2 presents the mathematical formulation of the AC Optimal Power Flow (AC-OPF) problem. Section 3 introduces the proposed PIHGNN methodology, including the graph-based representation of the power network and the design of the physics-informed loss function. Section 4 provides detailed validation results based on case studies conducted on the IEEE 39-bus and 118-bus systems. Finally, Section 5 concludes the paper and discusses the future research.

2. AC-OPF problem formulation

Under the assumption of three-phase balanced conditions and steady-state operational regime in power systems [30], the standard AC-OPF model can be formulated as follows:

$$\min_{\mathbf{P}_G,\mathbf{Q}_G,\mathbf{V},\theta} \sum_{i\in\mathcal{G}} C_i(P_{G,i}) \tag{1}$$

$$\text{s.t.} \quad P_i = V_i \sum_{j\in\mathcal{N}} V_j \left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) \tag{2}$$

$$Q_i = V_i \sum_{j\in\mathcal{N}} V_j \left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) \tag{3}$$

$$P_{ij} = V_i^2 G_{ij} - V_i V_j (G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) \tag{4}$$

$$Q_{ij} = -V_i^2 B_{ij} + V_i V_j (B_{ij}\cos\theta_{ij} - G_{ij}\sin\theta_{ij}) \tag{5}$$

$$P_{G,i}^{\min} \le P_{G,i} \le P_{G,i}^{\max}, \quad i \in \mathcal{G} \tag{6}$$

$$Q_{G,i}^{\min} \le Q_{G,i} \le Q_{G,i}^{\max}, \quad i \in \mathcal{G} \tag{7}$$

$$V_i^{\min} \le V_i \le V_i^{\max}, \quad i \in \mathcal{N} \tag{8}$$

$$S_{ij} \le S_{ij}^{\max}, \quad \forall (i,j) \in \mathcal{B} \tag{9}$$

where $P_{G,i}$ denotes active power generation at generator i and $P_{G,i}^{\max}$ and $P_{G,i}^{\min}$ denote their maximum and minimum outputs respectively. $C_i(\cdot)$ represents quadratic cost function $C_i = a_i P_{G,i}^2 + b_i P_{G,i} + c_i$. $\mathcal{G}$ is the generator set. V_i and θ_i are voltage magnitudes and phase angles for each bus. P_i and Q_i are active and reactive power injections at bus i. P_{ij} and Q_{ij} are active and reactive branch flows from bus i to bus j. V_i and V_j are the voltage magnitudes at buses i and j. $\theta_{ij} = \theta_i - \theta_j$ is the voltage phase angle difference. G_{ij} and B_{ij} are the conductance and susceptance of the line connecting buses i and j. $\mathcal{N}$ is the set of all buses in the system. S_{ij} is the apparent power flow on line (i,j), and $\mathcal{B}$ is the set of all transmission lines.

The AC-OPF problem presents three fundamental computational challenges. First, its inherent nonlinearity primarily stems from power flow equations containing trigonometric relationships, resulting in non-convex problem formulation. Second, the computational complexity escalates exponentially in large-scale power systems encompassing thousands of nodes and transmission lines, manifesting the curse of dimensionality. Third, dynamic fluctuations in renewable energy generation and load demand introduce stochastic uncertainties that substantially complicate real-time optimization processes. Overcoming these persistent challenges, the synergistic convergence of topology-aware graph neural networks and physics-informed learning offers a robust and efficient solution. Beyond a theoretical paradigm shift, the empirical results demonstrate a substantial leap in computational efficiency: the proposed method achieves speedups of up to ×218.85 while strictly maintaining solution feasibility. This provides a validated framework for the real-time resolution of security-constrained AC-OPF problems.

3. Physics-informed heterogeneous graph neural networks for AC-OPF learning

To address the three computational challenges summarized in Section 2, the proposed PIHGNN framework is built on three complementary design principles. First, to cope with the strong nonlinearity and non-convexity of AC-OPF, PIHGNN learns an offline surrogate mapping from operating conditions and network topology to the optimal system state. This surrogate is parameterized by a deep neural network and regularized via physics-informed loss terms, encouraging consistency with AC power-flow equations, bus-type setpoints, and operational limits, while enabling direct (non-iterative) inference at runtime and avoiding per-instance convergence failures typical of nonlinear iterative solvers. Second, to improve scalability and mitigate the curse of dimensionality, we adopt a heterogeneous graph architecture that explicitly leverages grid sparsity and topology (and distinguishes generator/load roles), with shared message-passing parameters that do not grow with the number of buses, and an inference cost that scales primarily with network connectivity. Finally, to handle stochasticity and real-time requirements, the

framework follows a "train-offline, infer-online" paradigm that shifts the computational burden to data generation and training, enabling sub-millisecond AC-OPF inference.

3.1. Heterogeneous graph construction

The power system is modeled as a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. The node set is divided into Generator nodes $\mathcal{V}_g$ (comprising PV buses and the slack bus) and Load nodes $\mathcal{V}_l$ (PQ buses).

Node feature encoding. The input features differentiate between generator nodes and load nodes based on their physical attributes and operational constraints. Specifically, for load nodes, the input feature vector x_l consists of active and reactive power demands, defined as $x_l = [p_l, q_l]^\top$. For generator nodes, to avoid data leakage and incorporate physical constraints, the input feature vector x_g is constructed using power generation limits and local load demands:

$$x_g = [p_g^{\max}, p_g^{\min}, q_g^{\max}, q_g^{\min}, p_{l,g}, q_{l,g}]^\top \tag{10}$$

where $p_g^{\max}, p_g^{\min}, q_g^{\max}, q_g^{\min}$ denote the upper and lower limits of active and reactive power generation, respectively, and $p_{l,g}, q_{l,g}$ represent the active and reactive load demands at the generator bus. These inputs are projected to high-dimensional embeddings $h^{(0)}$ using type-specific learnable linear layers:

$$h_g^{(0)} = W_g x_g + b_g, \quad \forall g \in \mathcal{V}_g \tag{11}$$

$$h_l^{(0)} = W_l x_l + b_l, \quad \forall l \in \mathcal{V}_l \tag{12}$$

where $W_g \in \mathbb{R}^{d\times 6}$ and $W_l \in \mathbb{R}^{d\times 2}$ are learnable weight matrices, and $b_g, b_l \in \mathbb{R}^d$ are bias vectors. The parameter d denotes the dimension of the unified latent embedding space, ensuring that heterogeneous inputs are mapped to feature vectors of the same size.

Edge feature encoding. Edges are classified into two types based on connectivity: *Transmit* edges (E_{gl}) connecting generators to load buses, and *Connect* edges (E_{ll}) connecting load buses directly. Direct connections between generator nodes are excluded as generators physically couple to the grid via transmission buses. The input edge attributes differ by type: Transmit edges incorporate transmission limits, defined as $x_{gl} = [R, X, B, \text{Rate}]^\top \in \mathbb{R}^4$, while Connect edges utilize impedance parameters $x_{ll} = [R, X, B]^\top \in \mathbb{R}^3$. These features are encoded into a unified latent space using type-specific MLPs ϕ_{trans} and ϕ_{conn}:

$$e_{ij} = \begin{cases} \phi_{trans}(x_{ij}), & \text{if } (i,j) \in E_{gl} \\ \phi_{conn}(x_{ij}), & \text{if } (i,j) \in E_{ll} \end{cases} \tag{13}$$

where $\phi_{trans} : \mathbb{R}^4 \to \mathbb{R}^d$ and $\phi_{conn} : \mathbb{R}^3 \to \mathbb{R}^d$ allow for heterogeneous feature processing.

3.2. Hierarchical message passing

To capture the grid topology, we employ a heterogeneous graph attention mechanism. We perform message passing separately for each connection type to allow the model to learn distinct interaction patterns. Let $h_i^{(k)}$ denote the hidden state of node i at layer k. The update rule for layer $k+1$ involves aggregating messages from different neighbor types.

Attention calculation. For a generic edge connecting node j to node i, the attention coefficient α_{ij} is computed as:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}_r^\top [W_r h_i^{(k)} || W_r h_j^{(k)} || e_{ij}]))}{\sum_{n\in\mathcal{N}_i} \exp(\text{LeakyReLU}(\mathbf{a}_r^\top [W_r h_i^{(k)} || W_r h_n^{(k)} || e_{in}]))} \tag{14}$$

where LeakyReLU denotes the LeakyReLU activation function with a negative slope of $\alpha = 0.2$, which is standard for GAT attention mechanisms. r denotes the specific edge relation type (e.g., 'transmits' or 'connects') between node j and node i. $W_r \in \mathbb{R}^{d\times d}$ is the learnable weight matrix specific to relation r, used to project node features into the attention subspace. Similarly, $\mathbf{a}_r \in \mathbb{R}^{2d+d_{\text{edge}}}$ is the learnable attention vector for relation r, where d_{edge} denotes the dimension of the edge feature vector e_{ij}. The symbol $||$ denotes the concatenation operation. This relation-specific parameterization allows the model to learn distinct interaction patterns for different physical couplings in the power grid.

Message aggregation. The aggregation step differentiates between neighbor types. For a generator node $g \in \mathcal{V}_g$, it only receives messages from connected load buses ($\mathcal{N}_g^{(l)}$) via the reverse transmission lines:

$$m_{gl} = \sum_{j\in\mathcal{N}_g^{(l)}} \alpha_{gj} W_{gl} h_j^{(k)} \tag{15}$$

For a load node $l \in \mathcal{V}_l$, we aggregate messages from neighboring generators ($\mathcal{N}_l^{(g)}$) and connected loads ($\mathcal{N}_l^{(l)}$):

$$m_{lg} = \sum_{j\in\mathcal{N}_l^{(g)}} \alpha_{lj} W_{lg} h_j^{(k)} \tag{16}$$

$$m_{ll} = \sum_{j\in\mathcal{N}_l^{(l)}} \alpha_{lj} W_{ll} h_j^{(k)} \tag{17}$$

where W_{gl}, W_{lg}, W_{ll} are transformation matrices specific to the edge relation type, and α_{gj}, α_{lj} are the attention coefficients calculated via Eq. (14).

Node update. The node states are updated by fusing the aggregated messages with residual connections:

$$h_g^{(k+1)} = \sigma_R\left(h_g^{(k)} + m_{gl}\right) \tag{18}$$

$$h_l^{(k+1)} = \sigma_R\left(h_l^{(k)} + m_{lg} + m_{ll}\right) \tag{19}$$

where $\sigma_R(\cdot)$ denotes the ReLU activation function. This formulation specifically accounts for the physical topology where generators inject power into the grid via buses.

3.3. State decoding

After K layers of message passing, we aggregate the node and edge embeddings to form a global graph representation Z_{global} via global mean pooling, as implemented in the provided code:

$$Z_{\text{global}} = \text{Concat}\left(\frac{1}{|\mathcal{V}_g|}\sum_{g\in\mathcal{V}_g} h_g^{(K)}, \frac{1}{|\mathcal{V}_l|}\sum_{l\in\mathcal{V}_l} h_l^{(K)}, \bar{e}_{\text{tr}}, \bar{e}_{\text{conn}}\right) \tag{20}$$

where $\bar{e}_{\text{tr}}$ and $\bar{e}_{\text{conn}}$ are the pooled features of transmission and connection edges, respectively.

This global feature vector is fed into a residual MLP decoder to predict the full system state vector. The decoder consists of residual blocks defined as:

$$z^{(k+1)} = \sigma_L\left(W_2^{(k)}\sigma_L(W_1^{(k)} z^{(k)}) + z^{(k)}\right) \tag{21}$$

where σ_L denotes the LeakyReLU activation with a negative slope of 0.1 to prevent dead neurons in the deep regression layers. The final output layer maps the latent representation to the voltage magnitudes $\hat{\mathbf{V}} \in \mathbb{R}^{|N|}$ and phase angles $\hat{\theta} \in \mathbb{R}^{|N|}$ for all buses, followed by physics-aware activations:

$$\hat{\mathbf{V}} = \sigma(\mathbf{z}_V), \quad \hat{\theta} = \tanh(\mathbf{z}_\theta) \tag{22}$$

where σ is the Sigmoid function and $\mathbf{z}_V, \mathbf{z}_\theta$ are the split outputs of the final linear layer.

3.4. Physics-informed loss function

The training objective is designed to ensure both high prediction accuracy and strict adherence to power system physics. The total loss function $\mathcal{L}$ is a weighted sum of a supervised learning term and physics-based penalty terms:

$$\mathcal{L} = \omega_{\text{MSE}}\mathcal{L}_{\text{MSE}} + \omega_{\text{slack}}\mathcal{L}_{\text{slack}} + \omega_{\text{pv}}\mathcal{L}_{\text{pv}} + \omega_{\text{pq}}\mathcal{L}_{\text{pq}} + \omega_{\text{ineq}}\mathcal{L}_{\text{ineq}} \tag{23}$$

where $\omega_{\text{MSE}}, \omega_{\text{slack}}, \omega_{\text{pv}}, \omega_{\text{pq}}$, and ω_{ineq} are hyperparameters balancing the supervised reconstruction error and specific physical constraints.

Supervised loss. We employ Mean Squared Error (MSE) to minimize the distance between the predicted state variables $(\hat{V}, \hat{\theta})$ and the ground truth labels provided by MATPOWER:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|\mathcal{N}|}\sum_{i\in\mathcal{N}}\left((\hat{V}_i - V_i^{\text{label}})^2 + (\hat{\theta}_i - \theta_i^{\text{label}})^2\right) \tag{24}$$

Physics-based equality constraints. To enforce the distinct physical roles of different bus types, we decompose the power flow equations into node-specific losses:

1) Slack Bus Loss ($\mathcal{L}_{slack}$): The slack bus (indexed as s) serves as the system reference. We strictly penalize deviations from its fixed voltage magnitude and zero phase angle reference:

$$\mathcal{L}_{\text{slack}} = (\hat{V}_s - V_{\text{ref}})^2 + (\hat{\theta}_s - 0)^2 \tag{25}$$

2) PV Node Loss ($\mathcal{L}_{pv}$): For generator buses (PV nodes), the voltage magnitude and active power injection must match the setpoints. We calculate the active power injection $P_i(\hat{V}, \hat{\theta})$ using the power flow (Eq. 2) and minimize:

$$\mathcal{L}_{\text{pv}} = \frac{1}{|\mathcal{V}_{\text{pv}}|}\sum_{i\in\mathcal{V}_{\text{pv}}}\left[(\hat{V}_i - V_i^{\text{set}})^2 + (P_i(\hat{V}, \hat{\theta}) - P_i^{\text{set}})^2\right] \tag{26}$$

3) PQ Node Loss ($\mathcal{L}_{pq}$): For load buses (PQ nodes), both active and reactive power injections are fixed. We enforce consistency between the calculated injections $P_i(\hat{V}, \hat{\theta}), Q_i(\hat{V}, \hat{\theta})$ and the specified load demand $P_{d,i}$ and $Q_{d,i}$:

$$\mathcal{L}_{\text{pq}} = \frac{1}{|\mathcal{V}_{\text{pq}}|}\sum_{i\in\mathcal{V}_{\text{pq}}}\left[(P_i(\hat{V}, \hat{\theta}) + P_{d,i})^2 + (Q_i(\hat{V}, \hat{\theta}) + Q_{d,i})^2\right] \tag{27}$$

Operational inequality constraints. We enforce operational limits using ReLU-based penalty functions. The inequality loss $\mathcal{L}_{\text{ineq}}$ aggregates violations for generator outputs (P_G, Q_G), voltage magnitudes (V), branch power flows (S_{ij}), and global power balance:

$$\begin{aligned}\mathcal{L}_{\text{ineq}} = &\sum_{i\in\mathcal{G}}\left[\text{ReLU}(P_{G,i} - P_{G,i}^{\max}) + \text{ReLU}(P_{G,i}^{\min} - P_{G,i})\right]^2 \\ &+ \sum_{i\in\mathcal{N}}\left[\text{ReLU}(\hat{V}_i - V_i^{\max}) + \text{ReLU}(V_i^{\min} - \hat{V}_i)\right]^2 \\ &+ \sum_{(i,j)\in\mathcal{B}}\text{ReLU}(S_{ij}(\hat{V}, \hat{\theta}) - S_{ij}^{\max})^2\end{aligned} \tag{28}$$

The overall structure of PIHGNN is shown in Fig. 1. First, the input node features are unified into high-dimensional embeddings through independent projection layers, while the edge features are converted into high-dimensional representations by an encoder; subsequently, three layers of heterogeneous graph convolutional (HeteroConv) layers deal with the different types of edge relationships in turn, each layer aggregates the neighbor information through the graph attention mechanism and introduces residual connections to superimpose shallow features to enhance the gradient propagation; finally, the node embeddings of the multilayered fusion of global topology and local interactions are mapped by the MLP to the predicted values of the voltage magnitude and phase angle, from which the values of power flow for each node can be obtained and the loss function is constructed.

4. Case study

4.1. Experiment setup

The effectiveness and performance of the proposed PIHGNN model are evaluated using the IEEE 39-bus and 118-bus test systems, while preserving their original topological and electrical parameters. The corresponding system characteristics are summarized in Table 1.

For each test system, we construct a dataset that spans diverse operating points by applying randomized load perturbations. Specifically, for each sample, the active and reactive demands (P_d, Q_d) at every load

Table 1
Parameters of IEEE test systems.

System	Slack Bus	P-V Bus	P-Q Bus	Branch
IEEE 39-bus	1	9	29	46
IEEE 118-bus	1	53	64	186

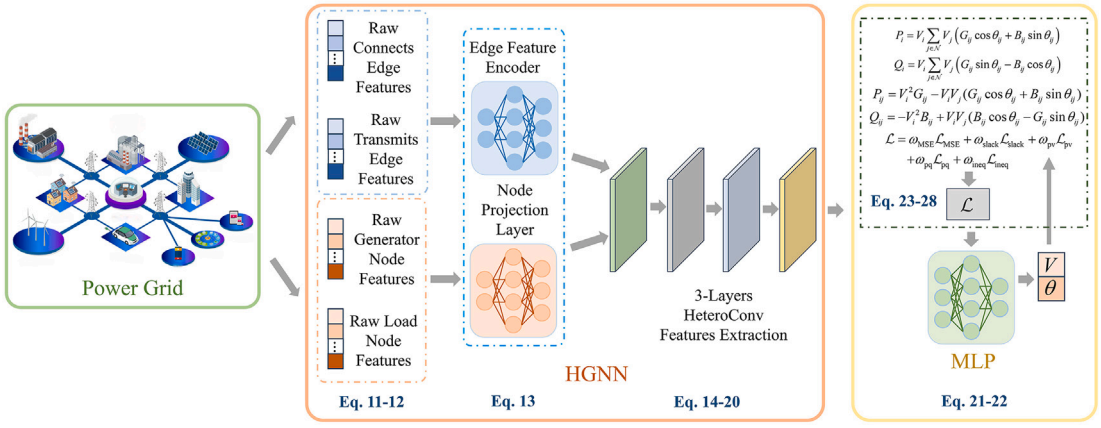


Fig. 1. The architecture of PIHGNN. The framework consists of Heterogeneous Graph Construction (Eqs. 11–13), Hierarchical Message Passing (Eqs. 14–20), and State Decoding (Eqs. 21 and 22), guided by the Physics-Informed Loss (Eqs. 23–28).

Table 2
Key model parameters configuration.

Category	Parameter	Value/Description
Optimizer	Algorithm	AdamW
	Weight Decay (L2)	1×10^{-4}
Training Config	Batch Size	32
	Epochs	1500
	Initial Learning Rate	3×10^{-4}
	EdgeDrop Rate	50%
LR Scheduler	Type	Cosine Annealing Warm Restarts
	T_0	10
GNN	Architecture	Heterogeneous GATConv
	Hidden Layers	3
	Hidden Dimension	64
	Feature Aggregation	Global Attention + Edge Fusion
MLP	Hidden Structure	512 → 256 → 128 + Residual Blocks
	Regularization	BatchNorm + Dropout (0.3 → 0.2)

(PQ) bus are independently drawn from a uniform distribution around their nominal values $(P_d^{\text{nom}}, Q_d^{\text{nom}})$ with a ±10% range:

$$P_d \sim \mathcal{U}\left(0.9\, P_d^{\text{nom}},\ 1.1\, P_d^{\text{nom}}\right), Q_d \sim \mathcal{U}\left(0.9\, Q_d^{\text{nom}},\ 1.1\, Q_d^{\text{nom}}\right). \tag{29}$$

This procedure yields a broad set of plausible loading conditions around the nominal operating point. For each sampled operating condition, an AC-OPF is solved using MATPOWER, and the resulting solution is treated as the reference (ground-truth) target for both training and evaluation. For each system, we generate 10,000 OPF samples and split the dataset into 80% training and 20% testing sets to assess generalization.

The hyperparameters used to train the proposed model are given in Table 2. We implement the proposed model on the PyTorch framework, employing the AdamW optimizer with integrated weight decay regularization for parameter training. The training configuration utilizes a batch size of 32 across 1500 epochs, with dynamic learning rate adjustment through a cosine annealed warm restart learning schedule. The model architecture comprises three principal components:

(1) A 3-layer heterogeneous graph attention network (GAT) with 64-dimensional hidden representations and residual connections for topological feature extraction;
(2) Multi-relational edge attribute integration through dedicated encoders for “transmits” (4-dim edge features) and “connects” (3-dim edge features) relationships;
(3) A deep MLP predictor with $512 \times 256 \times 128$ hidden layers using LeakyReLU activations with 0.1 negative slope, progressive dropout, batch normalization and adaptive output activation.

To improve robustness to topology variations and facilitate zero-shot generalization to N-1 contingencies, the EdgeDrop-based topology augmentation strategy is adopted during training. Concretely, each training sample is processed with probability 0.5 by masking (removing) exactly one edge. The masking is applied only to the transmits edges that connect generator nodes to load buses, thereby emulating a plausible transmission-line outage that perturbs generator-to-network coupling. By exposing the model to such randomized edge removals, PIHGNN is discouraged from exploiting spurious dependencies tied to a fixed N-0 topology and is instead encouraged to learn topology-robust representations and physically meaningful correlations. Consequently, the trained model can be directly evaluated on N-1 topologies at test time without additional fine-tuning, as demonstrated in the Results section.

The training and testing are conducted with an AMD EPYC 7763 64-Core Processor and 64GB RAM. We compare the proposed model with MATPOWER. To ensure a fair comparison, MATPOWER is tested on identical samples to those used by the PIHGNN model. Performance evaluation is conducted using the following four metrics: Convergence Rate (CR), Feasibility Rate (FR), Average Computing Time (ACT), and Average Cost Difference (δ_C):

$$\delta_C = \left| \frac{C_{\text{PIHGNN}} - C_{\text{MATPOWER}}}{C_{\text{MATPOWER}}} \right| \tag{30}$$

where C_{PIHGNN} and C_{MATPOWER} denote average cost of PIHGNN and MATPOWER respectively.

4.2. OPF results of PIHGNN

During the training process, the proposed model is updated once per epoch, with each update utilizing a batch of sample data. The loss value for each epoch is the average of the losses for all samples in that batch. The comprehensive evaluation of the PIHGNN model against MATPOWER demonstrates exceptional performance across multiple evaluation metrics. Table 3 presents a comparative analysis under uniformly sampled load scenarios with a ±10% fluctuation. The results reveal that both PIHGNN and MATPOWER achieve perfect convergence rates of 100% across both test systems, indicating robust optimization capability under varying load conditions.

In terms of feasibility rate, the PIHGNN model demonstrates perfect performance, achieving 100% for both the IEEE 39-bus and IEEE 118-bus systems as shown in Table 3. While MATPOWER also maintains a 100% feasibility rate, the results demonstrate that the proposed PIHGNN model produces solutions that satisfy AC-OPF constraints with absolute reliability, matching the performance of traditional iterative solvers.

PIHGNN delivers a substantial reduction in solution time compared with the conventional AC-OPF solver. As reported in Table 4 and illustrated in Fig. 2, the mean inference time is 0.612 ms for the IEEE 39-bus system and 0.645 ms for the IEEE 118-bus system. These values correspond to the amortized per-sample runtime measured under batched inference with batch size 32, reflecting high-throughput deployment settings. Fig. 2 uses a logarithmic axis to clearly contrast PIHGNN’s sub-millisecond inference with MATPOWER’s runtime, emphasizing the multi-order-of-magnitude gap. Beyond the mean, PIHGNN exhibits bounded tail latency: the standard deviation is 1.664 ms (39-bus) and 1.755 ms (118-bus), and the maximum observed inference time remains below 13.605 ms and 14.350 ms, respectively. In contrast, MATPOWER requires 102.944 ms (39-bus) and 141.160 ms (118-bus) on average, yielding speedups of ×168.21 and ×218.85 for the two systems.

This dramatic improvement in computational efficiency translates to significant practical advantages for real-time power system operations. The sub-millisecond response time of the PIHGNN model makes it particularly suitable for applications requiring rapid decision-making, such as real-time economic dispatch and emergency control scenarios.

We quantify economic optimality using the Average Cost Difference metric δ_C, defined as the relative generation-cost deviation of PIHGNN

Table 3
Comparison of convergence rate (CR) and feasibility rate (FR) between PIHGNN and MATPOWER (%).

System	CR		FR	
	PIHGNN	MATPOWER	PIHGNN	MATPOWER
IEEE 39-bus	100	100	100	100
IEEE 118-bus	100	100	100	100

Table 4
Statistical comparison of computational time (ms) and speedup between PIHGNN and MATPOWER.

System	Method	Computational Time (ms)				Speedup
		Mean	Std.	Min	Max	
IEEE 39-bus	PIHGNN	0.612	1.664	0.380	13.605	×168.21
	MATPOWER	102.944	49.381	88.327	2152.974	
IEEE 118-bus	PIHGNN	0.645	1.755	0.401	14.350	×218.85
	MATPOWER	141.160	46.483	122.660	2061.568	

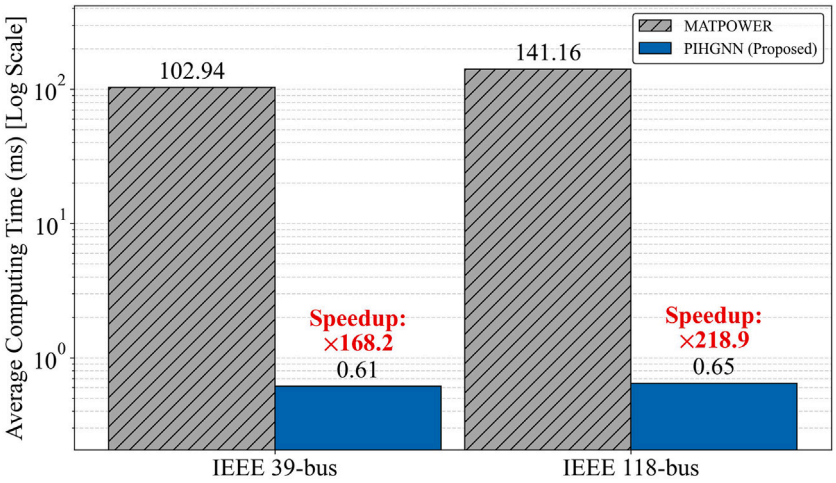


Fig. 2. Comparison of average computing time between PIHGNN and MATPOWER on a logarithmic scale. The red labels indicate the speedup factors achieved by the proposed model across different IEEE test systems. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5
Statistical analysis of cost difference (δ_C) for IEEE 39-bus and IEEE 118-bus systems.

System	Metric	Mean	95% CI	Std.	Min	Max
IEEE 39-bus	δ_C (%)	0.1276	[0.1232, 0.1320]	0.0999	0.0001	0.7409
IEEE 118-bus	δ_C (%)	0.0282	[0.0277, 0.0287]	0.0115	0.0050	0.1256

from the MATPOWER AC-OPF benchmark. As summarized in Table 5, PIHGNN achieves very small mean cost deviations of 0.1276% on the IEEE 39-bus system and 0.0282% on the IEEE 118-bus system. Moreover, the performance is statistically stable across the test set: the 95% confidence interval (CI) of the mean is tight (e.g., [0.0277, 0.0287]% for the IEEE 118-bus case), indicating low uncertainty in the estimated average deviation. Importantly, the worst-case cost deviation remains bounded, with a maximum δ_C of 0.7409% on the IEEE 39-bus system, suggesting that PIHGNN does not exhibit severe degradation even at challenging operating points.

The inverse relationship between system size and cost difference (lower δ_C for larger systems) suggests that the PIHGNN model's physics-informed architecture becomes increasingly effective as system complexity grows. This scalability characteristic is particularly valuable for large-scale power system applications, where traditional optimization methods often encounter computational bottlenecks.

To enable a rigorous and fair assessment of both latency and solution quality, Tables 4 and 5 report distributional statistics (e.g., mean, standard deviation, extrema, and confidence intervals where applicable), rather than only average values. MATPOWER serves as the exact AC-OPF benchmark but incurs substantially higher runtime. In contrast, PIHGNN produces near-optimal solutions with negligible economic impact (Max $\delta_C < 0.15\%$) while exhibiting tightly bounded inference-time variability. These results indicate that PIHGNN is not only faster on average, but also provides predictable, robust performance suitable for time-critical operational settings, making it a practical alternative to iterative solvers when low latency is a primary requirement.

4.3. Fast security-constrained OPF for N - 1 topology changes

The robustness of the proposed PIHGNN model under varying topologies is essential for its practical deployment in real-world power systems. To this end, this section evaluates the model's performance under N-1 topology changes, which represent the most common and critical scenarios considered in power system security analysis.

To evaluate the robustness of PIHGNN, we systematically simulate N-1 topology variations by removing selected transmission lines from the IEEE 39-bus and IEEE 118-bus test systems. For each contingency scenario, the model predicts the post-contingency optimal power flow without retraining or fine-tuning, utilizing only the modified network topology as input. The computation accuracy is assessed through the feasibility rate of the test results. The topological structures of each system are shown in Table 6.

The experimental results demonstrate exceptional robustness of the PIHGNN model across various N-1 topology change cases, as presented in Table 6. For the IEEE 39-bus system, the model achieves consistently high feasibility rates across all three contingency types, with values of 99.9%, 100%, and 100% for T1, T2, and T3 cases, respectively. The minimal variation in performance among different contingency types indicates that the model maintains stable computational capability even under diverse network disruption conditions.

Remarkably, the IEEE 118-bus system exhibits perfect feasibility rates of 100% across all three N-1 topology change cases (T1, T2, and T3). This outstanding performance on a more complex and larger-scale power system demonstrates the superior scalability and generalization capability of the PIHGNN architecture. The perfect feasibility rates suggest that the physics-informed design effectively captures the underlying power system constraints and maintains solution validity even when contingencies alter the network topology.

The comparison between the IEEE 39-bus and 118-bus test systems reveals that the model's robustness is maintained and even enhanced as the system complexity increases. While conventional neural network approaches often suffer from performance degradation in larger systems due to increased dimensionality, the PIHGNN model shows improved performance in the IEEE 118-bus system. This counterintuitive

Table 6
Performance and topology configuration of PIHGNN under N-1 contingency scenarios (T1, T2, T3).

System	Metric	Contingency scenario		
		T1	T2	T3
IEEE 39-bus	Removed Line	3–18	17–27	26–28
	Feasibility Rate (%)	99.9	100	100
IEEE 118-bus	Removed Line	4–5	23–25	65–68
	Feasibility Rate (%)	100	100	100

result can be attributed to the heterogeneous graph representation and physics-aware architecture, which better leverage the increased connectivity and redundancy present in larger power systems.

The consistently high feasibility rates across all scenarios demonstrate that the PIHGNN model generates practically viable solutions that adhere to operational constraints, underscoring its suitability for real-time power system applications where feasibility is paramount. Moreover, the minimal performance variation observed across cases T1, T2, and T3 further validates the model's robust generalization capability under diverse operational disturbances.

5. Conclusion

This study proposes a PIHGNN framework to accelerate the solution of AC-OPF problems while preserving solution fidelity and physical feasibility. By embedding operational constraints directly into the model architecture and loss function, the proposed approach synergistically combines data-driven learning with first-principles-based physical modeling. Extensive evaluations on the IEEE 39-bus and 118-bus test systems demonstrate the framework's capacity to achieve real-time performance, thereby offering a promising pathway toward scalable deployment in modern and future power system operations.

From a computational perspective, the PIHGNN framework achieves substantial acceleration, delivering speedups of ×168.21 and ×218.85 on the IEEE 39-bus and 118-bus systems, respectively, compared to conventional nonlinear OPF solvers such as MATPOWER. This significant reduction in computational latency, with average computing times of 0.612 ms and 0.645 ms for the respective systems, underscores the model's suitability for time-sensitive applications such as real-time economic dispatch, fast contingency screening, and rolling optimization under uncertainty. Notably, these computational gains are achieved without sacrificing solution quality: the resulting dispatch solutions exhibit cost deviations as low as 0.0282% compared to established benchmarks, validating the model's ability to approximate optimal solutions with high fidelity.

In terms of operational reliability, the PIHGNN demonstrates outstanding feasibility performance under both normal and contingency conditions. Under uniformly sampled load scenarios with ±10% fluctuation, the framework achieves perfect convergence rates of 100% and feasibility rates of 100% for both the IEEE 39-bus and 118-bus systems. The robustness analysis against N-1 contingencies further validates the model's practical applicability, with feasibility rates reaching 100% across most tested scenarios. These results affirm the framework's robustness and reliability under diverse grid scenarios, highlighting its suitability for practical deployment in real-world grid environments where system security is paramount.

The framework's superior scalability is empirically supported by the experimental results on the larger grid. Contrary to conventional learning-based methods that often degrade with size, the IEEE 118-bus system exhibits a lower average cost difference ($\delta_C = 0.0282\%$) compared to the smaller IEEE 39-bus system ($\delta_C = 0.1276\%$). Furthermore, the robustness of the architecture is validated by the perfect 100% feasibility rate achieved across all tested N-1 contingency scenarios (T1-T3) on the 118-bus system. These data points confirm that the heterogeneous graph structure effectively leverages the increased connectivity of larger networks, minimizing error and maintaining reliability as system complexity grows.

Collectively, these findings establish the PIHGNN framework as an effective and scalable solution for power system optimization, offering an optimal trade-off between computational efficiency, physical interpretability, and operational feasibility. The model's combination of speed, reliability, and domain-consistent representation makes it a strong candidate for integration into next-generation energy management systems (EMS), control centers, and autonomous grid platforms, particularly in scenarios that require rapid decision-making under uncertain and dynamic grid conditions.

Despite these encouraging results, several limitations of the present framework should be acknowledged. First, the model is data-dependent: its accuracy and reliability ultimately hinge on the coverage, fidelity, and scale of the offline training dataset, and performance may degrade under out-of-distribution operating conditions. Second, the physics-informed objective enforces network physics and operational limits through weighted soft penalty terms. Although the empirical feasibility rates reached 100% in our tests, this design does not provide a formal guarantee of strict satisfaction of the AC power-flow equality constraints (i.e., Kirchhoff-consistent power balance) or operational inequality constraints (e.g., voltage bounds and equipment limits) in all extreme corner cases. Third, while the model shows strong robustness to single-line (N-1) outages, more complex N-k contingencies or substantial topological reconfigurations may require additional data augmentation and/or fine-tuning to preserve near-optimal performance.

To address these limitations, future architectural developments should focus on more rigorously integrating hard physical and operational constraints within the learning process. Emerging hybrid paradigms that couple deep neural networks with differentiable optimization layers hold the potential to ensure stricter compliance with grid constraints. Additionally, incorporating online learning capabilities would enable real-time adaptation to changing operating conditions, while robustness to missing data and communication delays should be prioritized through uncertainty-aware learning techniques.

In conclusion, this study presents a scalable and efficient solution for optimal power flow that retains exceptional physical fidelity and decision reliability. By achieving substantial improvements in computation time while maintaining superior solution quality under both normal and contingency conditions, the approach opens the door to real-time, data-driven optimization in large-scale, uncertain, and dynamic environments.

CRediT authorship contribution statement

Liangcai Zhou: Supervision, Software, Methodology, Formal analysis, Conceptualization. **Pan Wu:** Visualization, Validation, Investigation, Formal analysis, Data curation. **Rui Chen:** Writing – original draft, Software, Investigation, Data curation. **Xin Chen:** Writing – review & editing, Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The raw data supporting the conclusions of this article will be made available by the authors upon request.

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